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Assignment 3

Trying Attacks

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1.

```
kadetj@ubuntuserver-lab:~/Desktop$ export MYVAR="hello"
kadetj@ubuntuserver-lab:~/Desktop$ printenv MYVAR
hello
kadetj@ubuntuserver-lab:~/Desktop$ unset MYVAR
kadetj@ubuntuserver-lab:~/Desktop$ printenv MYVAR
kadetj@ubuntuserver-lab:~/Desktop$
```

Made a variable with export, printed it, unset it to remove the content of that variable, and printed again to show there was no content anymore.

2.

```
kadetj@ubuntuserver-lab:~/Desktop$ gcc myprintenv.c
kadetj@ubuntuserver-lab:~/Desktop$ gcc myprintenv.c
kadetj@ubuntuserver-lab:~/Desktop$ ./a.out > child.txt
kadetj@ubuntuserver-lab:~/Desktop$ gcc myprintenv.c
kadetj@ubuntuserver-lab:~/Desktop$ ./a.out > parent.txt
kadetj@ubuntuserver-lab:~/Desktop$ diff child.txt parent.txt
kadetj@ubuntuserver-lab:~/Desktop$
```

There were no differences between the two files because the child inherits from the parent so without the parent, the child inherits nothing, and without the child there is nothing to inherit.

```
kadetj@ubuntuserver-lab:~/Desktop$ gcc myenv.c
kadetj@ubuntuserver-lab:~/Desktop$ ./a.out
kadetj@ubuntuserver-lab:~/Desktop$ gcc myenv.c
kadetj@ubuntuserver-lab:~/Desktop$ ./a.out
SHELL=/bin/bash
SESSION_MANAGER=local/ubuntuserver-lab:@/tmp/.ICE-unix/1681,unix/ubuntuserver-lab:/tmp/.ICE-unix/1681
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
SSH_AGENT_LAUNCHER=gnome-keyring
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/kadetj/Desktop
LOGNAME=kadetj
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=1709
XAUTHORITY=/run/user/1000/.mutter-Xwaylandauth.6AHOM3
HOME=/home/kadetj
USERNAME=kadetj
IM_CONFIG_PHASE=1
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;3
1:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*
.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war
=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=0
1;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.s
vg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35
:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.y
uv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.r
a=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
VTE_VERSION=6800
WAYLAND_DISPLAY=wayland-0
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/547049e1_7030_4cb4_b6ed_2bfe545140dd
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
```

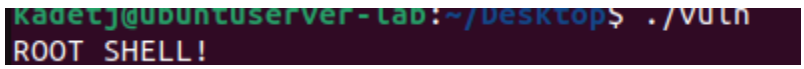
3. When NULL was in the argument, it doesn't allow the environment variables to pass because they are not automatically inherented by execve(), so you must make them pass with environ.

```
kadetj@ubuntuserver-lab:~/Desktop$ /bin/sh
$ gcc systemenv.c
$ ./a.out
LESSOPEN=| /usr/bin/lesspipe %s
USER=kadetj
XDG_SESSION_TYPE=wayland
SHLVL=1
HOME=/home/kadetj
OLDPWD=/home/kadetj
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
SYSTEMD_EXEC_PID=1709
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
COLORTERM=truecolor
IM_CONFIG_PHASE=1
WAYLAND_DISPLAY=wayland-0
LOGNAME=kadetj
_=systemenv.c
XDG_SESSION_CLASS=user
USERNAME=kadetj
TERM=xterm-256color
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
SESSION_MANAGER=local/ubuntuserver-lab:@/tmp/.ICE-unix/1681,unix/ubuntuserver-lab:/tmp/.ICE-unix/1681
XDG_MENU_PREFIX=gnome-
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/d14da0aa_b39a_4499_9ce0_58ec28f169d1
GNOME_SETUP_DISPLAY=:1
XDG_RUNTIME_DIR=/run/user/1000
DISPLAY=:0
LANG=en_US.UTF-8
Ubuntu-Server 22.04.5 LTS amd64
XDG_SESSION_DESKTOP=ubuntu
XAUTHORITY=/run/user/1000/.mutter-Xwaylandauth.6AHOM3
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.
lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:
*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=
01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng
=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=0
1;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*
.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
GNOME_TERMINAL_SERVICE=:1.196
SSH_AGENT_LAUNCHER=gnome-keyring
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
SHELL=/bin/bash
QT_ACCESSIBILITY=1
XDMSESSION=ubuntu
LESSCLOSE=/usr/bin/lesspipe %s %s
```

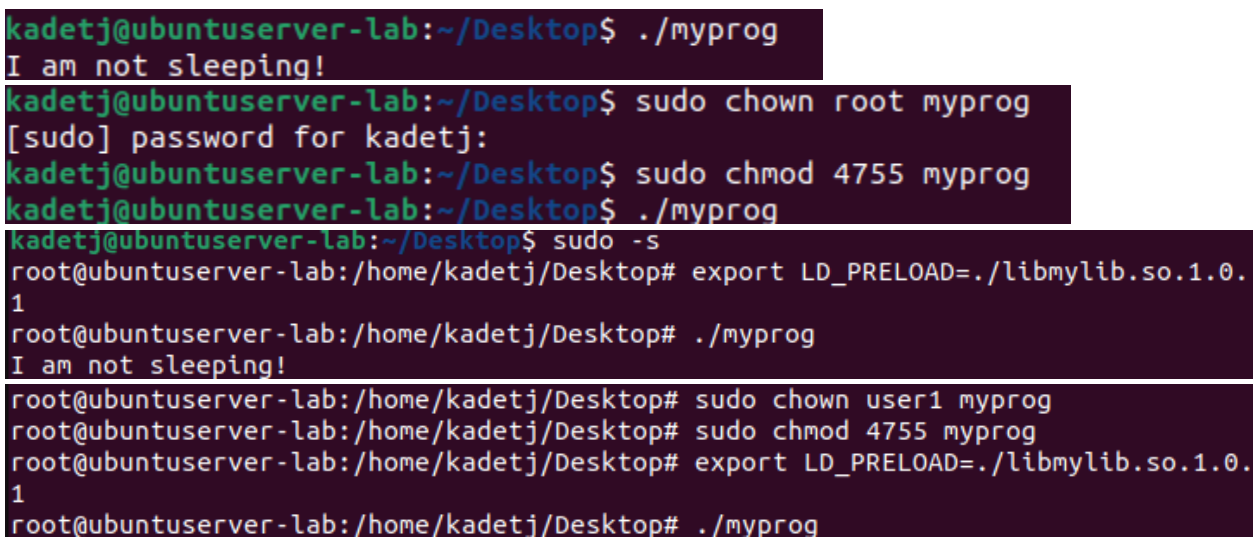
4. /bin/sh inherits the environment from the call.

5.

Path and MYVAR are inherited, and LD_LIBRARY_PATH isn't because Linux removes dangerous variables when you run a Set-UID.

6. 

Yes, my code ran instead of the /bin/ls. It being a Set-UID program causes it to run as a root, so yes, my program inherits those root privileges.

7. 

The first case is a normal program and not Set-UID, so it passes and runs the code. The second case is a Set-UID program, so it ignores privileged programs as security protection, so it doesn't pass. The third case passes because it is run as a root, so it is trusted. The fourth and final case does not run because it is a user making it a privileged program again, causing the security protection to kick in.

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
  printf("LD_PRELOAD = %s\n", getenv("LD_PRELOAD"));
  sleep(1);
  return 0;
}
```

After changing the program, case 2 and 4 now print as

```
root@ubuntuserver-lab:/home/kadetj/Desktop# ./myprog
LD_PRELOAD = (null)
```

The enviroment variable isnt loaded and it proves that it is cleaned due to it being a privileged program before it is run.

8.

```
kadetj@ubuntuserver-lab:~/Desktop$ ./catall "file.txt;rm /etc/zzz"
/bin/cat: file.txt: No such file or directory
rm: cannot remove '/etc/zzz': No such file or directory
```

Yes, Bob can compromise the system and delete a file because he can use argv[1] to write shell commands. As you can see, it attempts to delete a file.

```
kadetj@ubuntuserver-lab:~/Desktop$ ./catall "file.txt;rm /etc/zzz"
/bin/cat: 'file.txt;rm /etc/zzz': No such file or directory
```

When changed, it just searches for the file and does not attempt to remove it. It is not treated as a shell command like before.

9.

```
kadetj@ubuntuserver-lab:~/Desktop$ ./cap_leak
fd is 3
$ echo ' "
$ echo "HACKED" >&3
$ cat /etc/zzz
HACKED
```

You can now write to the root-owned file as a normal user. The shell remained open, allowing for commands to be entered. Privileges were dropped, but the privileged file descriptor remained open.